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REGULAR PAPER



Adversarially regularized medication recommendation model with multi-hop memory network

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Abstract
Medication recommendation is attracting enormous attention due to its promise in effectively prescribing medicines and improving the survival rate of patients. Among all challenges, drug–drug interactions (DDI) related to undesired duplication, antagonism, or alternation between drugs could lead to fatal side effects. Previous researches usually provide models with DDI knowledge to achieve DDI reduction. However, the mixed use of patients with different DDI rates places stringent requirements on the generalization performance of models. In pursuit of a more effective method, we propose the adversarially regularized model for medication recommendation (ARMR). Specifically, ARMR firstly models temporal information from medical records to obtain patient representations and builds a key-value memory network based on information from historical admissions. Then, ARMR carries out multi-hop reading on the memory network to recommend medications. Meanwhile, ARMR uses a GAN model to adversarially regulate the distribution of patient representations by matching the distribution to a desired Gaussian distribution to achieve DDI reduction. Comparative evaluations between ARMR and baselines show that ARMR outperforms all baselines in terms of medication recommendation, achieving DDI reduction regardless of numbers of DDI types being considered.

Keywords Medication recommendation · Drug–drug interactions · Memory networks · Adversarially regularization

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1 Introduction

Along with the development of medical and deep learning technologies, effective healthcare applications that cover from physiological condition monitoring and disease diagnosis to onset prediction are becoming more feasible [10,15,19,20,33,37]. The accumulation of electronic healthcare records (EHRs) [11], such as vital signs, laboratory tests, and clinical notes, provides researchers in both medical and deep learning fields with strong supporting evidence for analysis. Most of these applications attempt to assist physicians in better understandings of patient conditions and making accurate clinical decisions. As one of the most important applications, medication combination recommendation [28,35,40] enables physicians to make effective and safe prescriptions according to the medical information that describes the patient condition change, as well as expertise related to massive medications. The comprehensive analysis of patient information and medication knowledge could help to avoid the peril brought by the fast-paced way of working in intensive care units (ICU), significantly improving the survival rate of patients.

1.1 Motivation

Recommendation of medication combination is a challenging task for physicians due to the complex relations among patients, diseases, and medications. It is common for a patient to suffer from multiple diseases at the same time, resulting in the related EHRs informative and complex. In this case, it is difficult to obtain meaningful underlying patterns of patients for medication recommendation. Besides, advances in the pharmaceutical industry have provided doctors with more selections for different diseases. The knowledge gaps prevent them from making timely and accurate choices when facing massive drugs. More importantly, medications designed for different diseases may interact with each other, therefore leading to the reduction of effectiveness or even fatal side effects that people did not expect during the therapeutic period. Thus, relying on physicians' intuition and experience alone is not enough for making medication prescription; developing an effective and safe strategy for medication combination recommendation is imperative and extremely urgent.

1.2 Challenges

A series of approaches have been proposed for medication recommendation to address issues of the assessment of a patient's condition, the adverse interactions between medication, and the efficacy of medications. Assessing a patient's condition aims to derive clinically informative representations from EHRs and use them as the basis for medication recommendation. The two basic strategies for assessment are both instance-based method and longitudinal method [2,16,28], while the longitudinal method is preferred by most researchers since it makes good use of historical information for modeling representations [2,3,14,16]. Meanwhile, drug–drug interactions (DDI) are usually caused by duplication (two drugs with the same effect are taken), antagonism (two drugs with opposing actions), and alternation (one drug may alter the effect of another drug) [40]. Previous studies address the problem by graph augmented memory networks and reinforcement learning based on knowledge from the co-occurrence of medication in EHRs [17,28,40]. Exploring the impact of drugs on patients has always been the focus of researchers [1,4,21]. Traditionally, methods to test the effects of drugs include in vitro or in vivo laboratory analysis and computational models and simula-

tions (in silico) [29,30,38], while some researchers identify new relations between drugs and diseases in existing knowledge by mining academic publications [23,31,34].

However, patient have different DDI rates, i.e., given a medication set m, the fraction of combinations that would lead to DDI to the total number of combinations. All of those proposed methods treat patients with different DDI rates as a single whole when they feed models with auxiliary information to avoid DDI and assume that representations of those patients follow the same distribution. Therefore, the results of those methods are foreseeable: on the one hand, strong generalization is required by the model to eliminate the impact of patients with different DDI rates on the model performance; on the other hand, patients with different DDI rates may pose negative effects on each other when the model learning representations for them. Another drawback of those methods is that they read historical information only once for the assessment of patient conditions and ignore the fact that new evidence could be combined into patient representations and retrieve more pertinent information in subsequent accesses to previous admissions.

1.3 Solutions

Taking all these challenges into account, we propose adversarially regularized medication recommendation (ARMR). ARMR firstly builds an encoder that uses two GRU networks to obtain patient representations based on diagnoses and procedure. Then, ARMR develops a key-value memory network that stores historical representations and corresponding medications as pairs and carries out multi-hop reading on the memory network to obtain case-based similar information from historical EHRs, which is used to update patient representations. The encoder and memory network form a module called Medication Recommendation (MedRec). Meanwhile, ARMR combines the encoder as generator with a discriminator to form a GAN model, and representations of those patients whose DDI rates are smaller than a preset threshold are used as real data, so that the GAN model can shape the distribution of patient representations generated by the encoder to achieve DDI reduction. The MedRec module and GAN model are trained jointly within each mini-batch with dual objectives: a traditional error criterion for medication recommendation and an adversarial training criterion for distribution regulation. Such a strategy enables ARMR to learn meaningful patient representations and regulate data distribution to maintain low DDI simultaneously.

1.4 Contributions

Our contributions could be summarized as follows:

- A novel medication recommendation model that iteratively learns historical information of patients by multi-hop reading on the key-value memory network. Hence, ARMR keeps obtain new case-based similar information from previous admissions to update patient representations for medication recommendation.
- A novel adversarial regularization strategy that achieves DDI reduction by shaping the distribution of patient representations. Patients with different DDI rates are used for training simultaneously without posing adverse effects on each other or placing stringent requirements on the generalization performance.
- Experimental results on a real-world EHR dataset demonstrate that ARMR is capable of making efficient medication prediction with f1 to be 0.6559, and DDI is effectively avoided regardless the number of DDI types being considered.

2 Related work

2.1 Memory neural networks

The memory neural networks (MemNN) provide neural network models with an external memory component that could be read and written to remember facts from the past [8,39]. But the model is not easy to train via backpropagation and requires supervision at each layer of the network, so [32] presents a end-to-end variant, making it more generally applicable in realistic settings. All these approaches implement a single memory hop operation for each query, which may not retrieve enough information from the memory component. Thus, [13] proposes to carry out multiple episodes on the memory module, so that new evidence can be combined into the query to focus on and retrieve more information in subsequent accesses. [22] further extends the memory component into key-value paired memories, which gives greater flexibility for the practitioner to encode prior knowledge via non-trivial transforms between key and value. MemNN has also been used in healthcare problems. [25] introduces condensed memory neural networks (C-MemNNs), a model with iterative condensation of memory representations that preserves the hierarchy of features in the memory, while [28] uses MemNN to integrate DDI knowledge for safe medication recommendation.

2.2 Generative adversarial networks

The generative adversarial network (GAN) is an effective generative model initially designed for generating high-quality synthetic images [7], and the basic principle is to make a probability distribution to follow the distribution as the real data does. GAN is also used to regularize distribution when real data are absent. For instance, [24] combines an autoencoder (AE) with the GAN model for graph embedding, and uses the GAN model to make encoder outputs follow a standard Gaussian distribution. [12] makes use of a similar principle for de Novo generation of new molecules with desired properties. Both of these approaches arbitrarily set a single standard Gaussian distribution as a real data source with no prior knowledge. Meanwhile, [18] uses mixture Gaussian distribution to generate real data, incorporating the label information in the adversarial training stage to better shape the distribution of hidden states. Moreover, [6] and [9] add an extra regression task to AE for property prediction so that the encoder outputs contain information about desired properties, which could be used for adversarial training. These proposed methods either use arbitrary distributions that could not reflect desired properties or add extra tasks to shape distributions. In contrast, our method directly fits the probability distribution that covers the desired properties to generate real data without extra tasks.

2.3 Medication recommendation

The recommendation of medication combination has received enormous attention in recent years. [14] uses memory augmented neural networks to model the complex interactions between different sequential views in the EHR, aiming to achieve better performance in the tasks of drug prescription and disease progression in healthcare. But the model did not take drug–drug interaction into account, so [28] integrates the drug–drug interactions knowledge graph by a memory module and models the longitudinal patient records as the query. Meanwhile, [40] formulates treatment recommendations as a sequential decision-making problem and uses a recurrent decoder to model drug dependency, content-based attention to model the

underlying drug-to-disease mapping. To model both visit-level and variable-level sequential information in the EHR, [3] presents a two-level neural attention model that detect influential past visits and significant clinical variables within those visits, which could provide a detailed interpretation of the prediction results while retaining the prediction accuracy. Meanwhile, important data fields for predicting medications could be determined through interviews with pharmacists, and they are used to display the most probable medications on top of the drop-down list in the e-prescription application [5]. Most of these works have ignored longitudinal EHRs related to patients with a single visit, so [27] pre-trains its transformer-based visit encoder on EHR data from patients with a single visit, and then, the model is fine-tuned on EHRs from patients with multiple visits.

Compared to previous methods, ARMR obtains temporal information using GRU from historical admissions and uses it to develop a key-value memory network, where patient representations and corresponding medications are stored as keys and values. Moreover, ARMR carries out multi-hop reading on the memory network based on attention mechanism to obtain case-based similar information. Such an operation enables ARMR to infer on historical admissions and acquire new information when the representation of the current admission is updated. ARMR also uses the GAN model to regulate distributions of patient representations, so that these representations would be shaped to show low DDI rates to achieve DDI reduction.

3 Proposed method

In this section, we describe the details of ARMR. ARMR firstly uses RNNs as encoder to obtain patient representations and constructs a key-value memory network that stores historical representations and corresponding medications, so that case-based similar information could be used for medication recommendation. We call the module that contains the encoder and memory network as MedRec. Then, ARMR uses a GAN model to match distribution of patient representations to a prior Gaussian distribution to achieve DDI reduction. The MedRec module and GAN model are trained with dual objectives alternately within a mini-batch. The overview of the framework is illustrated in Fig. 1.

3.1 Notations

EHRs consist of multiple types of longitudinal information that describe patients in terms of different views, so each patient could be viewed as a sequence of multivariate observations $X = [x_1, x_2, \ldots, x_T]$, where T is the number of admissions and x_t is the representation of the tth visit. Each visit x_t contains three sets of medical codes that represent corresponding diagnoses d^t, procedures p^t, and medications m^t, and the total number of diagnoses, procedures, and medications are $|C_d|$, $|C_p|$ and $|C_m|$, respectively. d^t, p^t, and m^t are transformed into multi-hot vectors $c_d^t \in \mathbb{R}^{|C_d|}$, $c_p^t \in \mathbb{R}^{|C_p|}$, and $c_m^t \in \mathbb{R}^{|C_m|}$.

To indicate DDI between medications, a DDI matrix $DM \in \mathbb{R}^{|C_m|*|C_m|}$ is constructed, and $DM_{i,j} = DM_{j,i} = 1$ if medication i and j would lead to DDI or $DM_{i,j} = DM_{j,i} = 0$ if not. Thus, given DM, the DDI rate Dr_{m^t} of m^t could be calculated according to Eq. (1), i.e., the fraction of combinations that would lead to DDI to the total number of combinations,

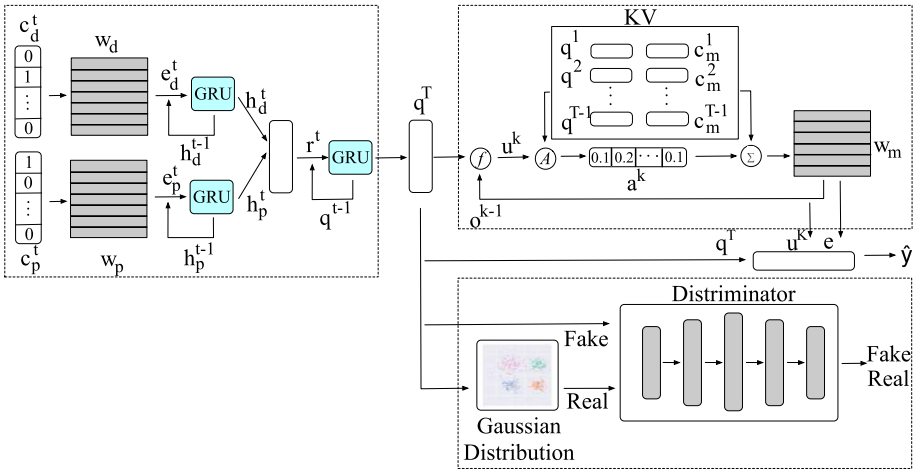


Fig. 1 The workflow of the proposed ARMR model. For the tth visit, the diagnoses c_d^t and procedures c_p^t are transformed into embedding vectors e_d^t and e_p^t by embedding matrices W_d and W_p and fed into two separate RNNs. Then, h_d^t and h_p^t are combined by a linear embedding layer to get the representation r^t, which would be processed by a separate GRU unit to get the final representation q^t. Then, all q^t ($t \in [1, T-1]$) are used to construct the key-value memory network, whose keys are historical representations q^t and values are corresponding medications c_m^t. Meantime, q^T is used to fit Gaussian distribution, which would provide the GAN model with real data while the fake data are generated by the encoder

and $Dr_{m^*} \in [0, 1]$.

$$\text{DDI rate} = \frac{\sum_{t=1}^{N} \sum_{i,j} |(m_i^t, m_j^t) \in m^t \; and \; DM_{i,j} = 1|}{\sum_{t=1}^{N} \sum_{i,j} 1} \tag{1}$$

Meanwhile, a threshold ε_{rec} for DDI rate is required, and patients whose Dr_{m^T} of the last admission is smaller than ε_{rec} are used to train MedRec, and their latent features q^T generated by the encoder would be used as real data for the GAN model. Specifically, we use the Gaussian distribution $p(z)$ that q^T follows as the source of real data.

Given the historical EHR of a patient $X = [x_1, x_2, \ldots, x_{t-1}]$ as well as the diagnoses d^T and procedures p^T of the last visit, the goal of medication recommendation is to predict the medications m^T.

3.2 Encoding patient representation

We encode historical diagnoses and procedures with two separate GRUs, and combine the hidden states with a GRU unit to obtain patient representations. At each time stamp of the encoding phrase, d^t and p^t are firstly transformed into multi-hot vectors c_d^t and c_p^t to indicate the occurrence of corresponding diagnoses and procedures, then they are embedded with $W_d \in \mathbb{R}^{|C_d| \times dim}$ and $W_p \in \mathbb{R}^{|C_p| \times dim}$ separately according to Eqs. (2) and (3), where dim is the embedding size.

$$e_d^t = W_d c_d^t \tag{2}$$

$$e_p^t = W_p c_p^t \tag{3}$$

The GRUs take embeddings $[e_d^1, e_d^2, \ldots, e_d^t]$ and $[e_p^1, e_p^2, \ldots, e_p^t]$ as inputs and generate hidden states h_d^t and h_p^t according to Eqs. (4) and (5). The representation q^t is then generated using h_d^t and h_p^t as Eqs. (6) and (7) show. Here, $[\cdot]$ is a concatenation operation, and $f(\cdot)$ represents a single layer fully connected network.

$$h_d^t = GRU_d(e_d^1, e_d^2, \ldots, e_d^t) \tag{4}$$

$$h_p^t = GRU_p(e_p^1, e_p^2, \ldots, e_p^t) \tag{5}$$

$$r^t = f([h_d^t, h_p^t]) \tag{6}$$

$$q^t = GRU(r^1, r^2, \ldots, r^t) \tag{7}$$

3.3 Medication recommendation

We construct a key-value memory network KV and conduct multi-hop reading on it to leverage historical information of patients. The encoded representation of patients, as well as the information read from KV, will be fed into the prediction layer to predict multi-hot vectors for medication recommendation.

Following the typical way of describing a memory network, four components are introduced [39]:

I: Input Feature Map converts the incoming input to the internal feature representation, which is conducted according to Eqs. (2)–(7).

G: Generalization generates and updates the memory representation, and a key-value memory network KV is used here to model historical information. For a patient with T admissions, all historical representations $q^t (t \in [1, T-1])$ obtained by the encoder are used as keys, while the corresponding medications c_m^t will act as values, here c_m^t is a multi-hot vector that indicate the ground truth medications for the tth admission.

O: Output Feature Map carries out multi-hop reading on KV to extract information from historical admissions. Specifically, the attention weight a^k for the kth hop is calculated according to the general attention Eq. (8), where u^{k-1} is the output from the last hop, W_a is a learnable weight matrix for the general attention, and q^* are keys of KV. The attention a^k is then further used to read weighted memories from KV as Eq. (9) shows, where c_m^* refer to all the values in KV and $W_m \in \mathbb{R}^{|C_m| \times dim}$ is the embedding matrix that transforms the weighted sum of values to a dense embedding vector. Meanwhile, the query u^{k-1} is updated according to Eq. (10). The whole reading and updating process is repeated for K hops, while W_m is also updated by back-propagation along the training process, and q^T would act as u^0 at the very beginning of the multi-hop reading. Moreover, after K hops of reading and updating, q^T is used to retrieve the correlation between the representation of a patient and different medications based on standard attention as Eq. (11) shows, where W_e is a learnable weight matrix for the standard attention.

$$a^k = \text{Softmax}((u^{k-1})^\intercal W_a q^t) \quad t \in [1, T-1] \tag{8}$$

$$o^k = W_m c_m^* a^k \tag{9}$$

$$u^k = u^{k-1} + o^k \tag{10}$$

$$e = \text{Softmax}((q^T)^\intercal W_e W_m) W_m \tag{11}$$

The motivation for multi-hop reading on key-value memory network is that new evidence can be combined into the query to focus on and retrieve more pertinent information in subsequent access. Since we use historical records of a patient to construct the key-value memory

network, such a multi-hop reading on the network could take similar patient representation into consideration and make use of the corresponding medications for the prediction as Eqs. (8), (9), and (10) have shown. Meanwhile, the attention-based reading procedure shown by Eq. (11) helps to obtain the relation between the final patient representation and embeddings of medications.

R: Response converts the output into the response format desired. In the medication recommendation problem, the query q^T, the result of multi-hop u^K, and the weighted embeddings of medications e are fed into the prediction layer to predict medications as Eq. (12) shows, where $\sigma(.)$ is the sigmoid function and $f(.)$ is the fully connected layer. Then, we get the predict result in probability $\hat{y} \in \mathbb{R}^{|C_m|}$.

$$\hat{y} = \sigma(f([q^T, u^K, e])) \tag{12}$$

3.4 GAN regularization

We use the GAN model to regulate the distribution of patient representations generated by the encoder to avoid DDI, and the basic assumption for such a strategy is that different distributions reflect specific properties, including DDI, so given the distribution $p(z)$ that patients whose DDI rates are smaller than a preset threshold ϵ_{rec} follows, if all representations are regulated to follow $p(z)$, they would present the same property, i.e., low DDI rate.

Given the threshold ε_{rec}, all patients whose DDI rate $Dr_{m^T} \leq \varepsilon_{\text{rec}}$ are used to train the MedRec module described in Sects. 3.2 and 3.3 to get representations q_l^T, and we use the Gaussian distribution $p(z)$ that q_l^T follows as the source of real data. Specifically, we conduct Box-Cox transformation [36], which is commonly used to transform non-normal data to normal data, on q_l^T to get transformed data z, and take the mean of z as the mean vector of Gaussian distribution while an identical matrix is used as the covariance matrix. When all patients are used to train the MedRec module, the representations would be q_a^T, and the encoding distribution is $p(q_a^T|c_d^*, c_p^*)$. Here, q_a^T contains two types of data: q_l^T that represents patients whose $Dr_{m^T} \leq \varepsilon_{\text{rec}}$ and q_f^T that represents patients whose $Dr_{m^T} \in (\varepsilon_{\text{rec}}, 1]$.

The MedRec module is adversarially regularized by matching $p(q_a^T|c_d^*, c_p^*)$ to $p(z)$, so that the fake data, i.e., q_f^T, would be regulated to follow the same distribution $p(z)$ that the real data, i.e., q_l^T, follows. In order to do so, a discriminator is attached on top of the representations generated by the encoder, which would try to guide $p(q_a^T|c_d^*, c_p^*)$ to match $p(z)$. Meanwhile, MedRec attempts to minimize the error for medication recommendation, and the encoder is also the generator of the GAN model. The encoder ensures the encoding distribution $p(q_a^T|c_d^*, c_p^*)$ can fool the discriminator into thinking that the representations q_a^T come from the desired Gaussian distribution $p(z)$.

The GAN model and MedRec are trained jointly within each mini-batch in two phases: the prediction phase and the regularization phase.

In the prediction phase, the MedRec module is updated to minimize the weighted loss for medication recommendation shown in Eq. (13), which consists of the binary entropy loss $\mathcal{L}_{\text{bce}}$ and multi-label margin loss $\mathcal{L}_{\text{margin}}$. We use $\mathcal{L}_{\text{margin}}$ to make the predicated probability of ground truth medications has at least 1 margin larger than others, which would benefit the predictive performance of ARMR. Here, $y \in \mathbb{R}^{|C_m|}$ is c_m^T, i.e., the ground truth of medications for the final admission, and $\hat{y} \in \mathbb{R}^{|C_m|}$,$\hat{Y}$ are the predict results in the form of probability and medication set, respectively, so $\hat{y}_i$ and $\hat{y}[\hat{Y}_j]$ are the probabilities that the ith and $\hat{Y}_j$th medication would be prescribed to the patient. The constraints for the weights α_1 and α_2 are that $\alpha_1 > 0$, $\alpha_2 > 0$ and $\alpha_1 + \alpha_2 = 1$.

In the regulation phase, the GAN model firstly updates the discriminator to tell real data (samples from $p(z)$) from fake data q_f^T that generated by the encoder, then the generator, which is also the encoder of MedRec, is updated to confuse the discriminator, and the cost function for GAN regularization is the typical equation of the GAN model as shown in Eq. (16).

$$\mathcal{L} = \alpha_1 \mathcal{L}_{\text{bce}} + \alpha_2 \mathcal{L}_{\text{margin}} \tag{13}$$

$$\mathcal{L}_{\text{bce}} = -\sum_{i=1}^{|C_m|} [y_i \log \sigma(\hat{y}_i) + (1 - y_i) log(1 - \sigma(\hat{y}_i))] \tag{14}$$

$$\mathcal{L}_{\text{margin}} = \sum_{i=1}^{|C_m|} \sum_{j=1}^{|\hat{Y}|} \frac{\max(0, 1 - (\hat{y}[\hat{Y}_j] - \hat{y}[i]))}{L} \tag{15}$$

$$\min_{\mathcal{G}} \max_{\mathcal{D}} \mathbb{E}_{z \sim p(z)}[\log \mathcal{D}(\mathbf{Z})] + \mathbb{E}_{X \sim p(X)}[\log(1 - \mathcal{D}(\mathcal{G}(X)))] \tag{16}$$

Once the training procedure is done, patient representations q_a^T are regulated to follow $p(z)$, and these representations could still be mapped to corresponding medications to finish the recommendation. Thus, both the expected property, i.e., low DDI rate, and prediction performance are guaranteed.

The training progress is described by Algorithms 1, 2, and 3. We use Gaussian process to optimize hyper-parameters, and values of these hyper-parameters are shown in Table 1.

Algorithm 1 Recommending Medications

Require: EHRs x of a patient; number T of admissions; number of hops K on KV;
for t=1 to T **do**
  embed diagnoses and procedures to get e_d^t and e_p^t as Eq.(2) and Eq.(3) show
  encode medical records to get h_d^t and h_p^t as Eq.(4) and Eq.(5) show
  obtain patient representation q^t using Eq.(7)
end for
use q^t and $c_m^t (t \in [1, T-1])$ to construct KV
use q^T as u^0
for k=1 to K **do**
  read o^k from KV using Eq.(9)
  update u^{k-1} to get u^k according to Eq.(10)
end for
read from medication embeddings to get e according to Eq.(11)
predict medication $\hat{y}$ using Eq.(12)

4 Experiments

In this section, we conduct experiments on a real-world medical dataset MIMIC III to compare ARMR with different baselines. All experiments are conducted based on a NVIDIA TITAN Xp GPU, and the source code could be found at Github.[1]

[1] https://github.com/yanda-wang/ARMR.

Algorithm 2 Training MedRec

Require: EHRs X of all patients; training epoch E_1; batch size b; number of hops K on KV; DDI rate ε_{rec} for fitting distribution;

```
obtain records X_rec related to patients with Dr_{m^T} ≤ ε_rec;
for e=1 to E_1 do
    for i=1 to |X_rec|/b do
        call Algorithm 1 to get predict result ŷ and patient representation q^T
        use ŷ and ground truth y to train MedRec according to Eq.(13)
    end for
end for
fit Gaussian distribution p(z) using q^T
```

Algorithm 3 Training ARMR

Require: EHRs X of all patients; training epoch E_2; batch size b; Gaussian distribution $p(z)$; multi-hop reading K on KV; GAN regularization times R;

```
for e=1 to E_2 do
    for i=1 to |X|/b do
        call Algorithm 1 to get predict result ŷ
        use ŷ and ground truth y to train MedRec according to Eq.(13)
        for j=1 to R do
            sample b vectors {a_1, a_2, ..., a_b} from p(z) as real data
            use encoder to generate b representation vectors {q_1^T, q_2^T, ..., q_b^T} as fake data
            use the real and fake data to train the GAN model according to Eq.(16)
        end for
    end for
end for
```

Table 1 Values of hyper-parameters

Parameters	Values	Description
α_1	0.9	Weight of binary entropy loss $\mathcal{L}_{\text{bce}}$ in Eq. (13)
α_2	0.1	Weight of multi-label margin loss $\mathcal{L}_{\text{margin}}$ in Eq. (15)
batch_size	8	Batch size
en_dim	64	Dimension of hidden layers of GRU
en_n_layer	2	Number of hidden layers of GRU in the encoder
en_lr	3.994e−4	Learning rate of the optimizer for the encoder
hop	7	Number of multi-hop reading on KV
de_lr	7.882e−5	Learning rate of the optimizer for the decoder
de_dim	64	Dimension of medication representation in Eq. (9)
dis_lr	2.915e−5	Learning rate of the optimizer for the discriminator
dis_dim_B	143	Parameter for mini-batch discrimination [26]
dis_dim_C	55	Parameter for mini-batch discrimination

4.1 Dataset

MIMIC III v1.4 [11] is a publicly available dataset that consists of multiple types of medical information related to over 40,000 patients, which includes diagnoses, procedures, and prescriptions. We choose the set of medications prescribed by doctors during the first 24-h since the first 24-h is the most critical time for patients to obtain correct treatment quickly [28]. Moreover, we use the top-100 severe DDI types to calculate the DDI rate, and the probability

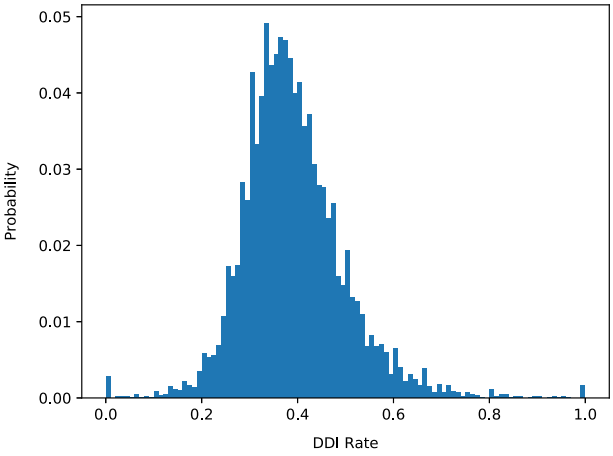


Fig. 2 Probability density of DDI rate

Table 2 Statistics of the dataset

Item	Train	Test	Validation
#patient	5080	635	635
avg#visit	2.0951	2.1216	2.2001
avg#diagnosis	14.0228	14.1237	14.2063
avg#procedure	4.3303	4.2074	4.4090
avg#medication	19.8573	19.7728	20.2111
avg DDI rate	0.3956	0.3952	0.3923

density of DDI rates of all patients is shown in Fig. 2. As Fig. 2 shows, the probability density approximates a Gaussian distribution with a mean of 0.4, so we set $\varepsilon_{\text{rec}} = 0.4$. Patients are divided into three parts with 80% for training, 10% for testing, and 10% for validation, and statistics of those datasets are summarized in Table 2.

4.2 Baselines

ARMR is compared to the following baselines, whose source codes are available in original papers:

- **Leap** [40]: Leap decomposes the treatment recommendation into a sequential decision-making process and reinforcement learning is leveraged to ensure accuracy and completeness.
- **RETAIN** [3]: RETAIN is designed as an interpretable predictive model for healthcare, which is based on a two-level neural attention model that detects influential past visits and significant clinical variables within those visits.
- **DMNC** [14]: DMNC treats medication prediction as a multi-view sequential problem and models view interactions as well as long-term dependencies via memory augmented neural network.

– **GAMENet** [28]: GAMENet takes patient history and DDI into account for medication recommendation via dynamic memory network and EHR graphs derived from longitudinal patient records.
– **MedRec[Section** 3]: MedRec is formed by the encoder and key-value memory network in ARMR for medication recommendation, but it is not regularized by the GAN model.

4.3 Metrics

To compare the performance of ARMR to those baselines, we use Jaccard similarity (Jaccard), precision recall AUC (PRAUC), and average F1 (F1) as metrics. Meanwhile, we also use the DDI rate and changes in the DDI rate (△DDI rate) to show whether ARMR successfully achieves DDI reduction.

Jaccard similarity is defined as the ratio of intersection of ground truth Y^t and predict result $\hat{Y}^t$ to the union of Y^t and $\hat{Y}^t$ as shown in Eq. (17), where N is the total number of patients.

$$\text{Jaccard similarity} = \frac{1}{\sum_{t=1}^{N} 1} \sum_{t=1}^{N} \frac{|Y^t \cap \hat{Y}^t|}{|Y^t \cup \hat{Y}^t|} \tag{17}$$

The metric F1 is defined as Eq. (18) shows, where Pre and $Recall$ are the precision and recall, which are shown in Eqs. (19) and (20), respectively.

$$F1 = \frac{1}{\sum_{t=1}^{N} 1} \sum_{t=1}^{N} \frac{2 * \text{Pre}^t * \text{Recall}^t}{\text{Pre}^t + \text{Recall}^t} \tag{18}$$

$$\text{Pre}^t = \frac{|Y^t \cap \hat{Y}^t|}{|\hat{Y}^t|} \tag{19}$$

$$\text{Recall}^t = \frac{|Y^t \cap \hat{Y}^t|}{|Y^t|} \tag{20}$$

To verify that ARMR is able to reduce DDI, we measure the change of DDI rate in the predicted results compared to the DDI rate in the ground truth as Eq. (21) shows.

$$\triangle \text{ DDI rate} = \frac{\text{DDI rate}(\hat{Y}) - \text{DDI rate}(Y)}{\text{DDIrate}(Y)} \tag{21}$$

4.4 Evaluation

Experimental results are shown in Table 3, and as we can see, ARMR outperforms all the baselines in terms of different metrics. ARMR achieves the lowest DDI rate as well as the only △DDI rate with a negative value, which indicate that it has reached the original intention of reducing DDI. MedRec, the variant of ARMR without a GAN model for adversarial regulation, achieves a △DDI rate of 1.22%, failing to reduce DDI rate. The comparison between ARMR and MedRec shows that the GAN model successfully regulates the distributions of patient representations to reduce DDI rate. GAMENet is another method that takes DDI knowledge into account, although GAMENet can achieve a lower DDI rate than those methods that do not consider DDI, its effect of reducing DDI is not as good as ARMR, which proves that regulating distribution of patient representations could be a more efficient strategy than treating DDI information as an extra source of knowledge to the model. All the other

Table 3 Comparison of different methods

Methods	DDI rate	△DDI rate (%)	Jaccard	PRAUC	F1
Leap	0.4229	7.80	0.4484	0.6457	0.6109
DMNC	0.4155	5.93	0.4933	0.7269	0.6511
RETAIN	0.4332	10.43	0.4897	0.7499	0.6494
GAMENet	0.4043	3.06	0.4970	0.7589	0.6544
MedRec	0.3971	1.22	0.4992	0.7535	0.6558
ARMR	0.3917	-0.15	0.5026	0.7613	0.6559

three baselines are not able to recommend safe medication combinations since they do not take DDI knowledge into consideration.

In addition to DDI reduction, ARMR also achieves better performance on medication recommendation, whose Jaccard, PRAUC, and F1 are higher than those of all baselines. Among these baselines, DMNC uses an encoder similar to ARMR, but its decoder does not consist of a key-value memory network or carry out multi-hop reading. The difference in performance between ARMR and DMNC indicates that key-value memory network and the multi-hop reading play an important role in effectively obtain meaningful information from similar historical admissions.

We perform kernel density estimates to compare DDI rates in the recommended results with the DDI rates in the real data, and the results are illustrated in Fig. 3. As shown in Fig. 3a, b, the density related to recommended results of Leap and DMNC is shifted to the right relative to the density of the real data, and this shift to high DDI rates explains why these two methods do not perform well in terms of △DDI rate. Meantime, Fig. 3c shows that although RETAIN also has a high △DDI rate, it is obtained in a different way since the density of recommended results has high values at high DDI rates with small variance, and GAMENet has a similar results shown in Fig. 3d, but the density shifts relatively little compared to high DDI rates. As for MedRec, Fig. 3e shows a similar result from Fig. 3d, except that the density is higher near 0.4. Finally, Fig. 3f shows that ARMR further optimizes the result by increasing density around 0.4, and the comparison with the result of MedRec tells us that ARMR also decreases the density around 0.5 to 0.6, proving that the GAN-based regularization is able to shape the representations of patients with high DDI rate to achieve DDI reduction.

Experiments are also conducted to explore how ARMR as well as those baselines are affected when different numbers of DDI types are considered, and the top-40, top-60, top-80, and top-100 severe DDI types are used, respectively, to calculate DDI rate. Results are shown in Table 4. As we can see, ARMR is the only model that is able to achieve DDI reduction when the number of DDI types being considered changes from 40 to 100, although △DDI rate climbs from -5.17 to -0.15%. Meanwhile, MedRec fails to reduce DDI rate when adversarial regulation is removed, and △DDI rate is always above zero regardless of the number of DDI types. GAMENet is another method that makes an effort to reduce DDI rate, and it succeeds when top-40 DDI types are used, achieving a △DDI rate of -1.08%, but when the number of DDI types increases from 40 to 100, GAMENet is not able to address DDI any more. As for the rest baselines, △DDI rate of Leap grows dramatically from -17.80 to 7.80% as DDI types being considered increases from 40 to 100, while neither DMNC nor RETAIN is able to achieve DDI reduction.

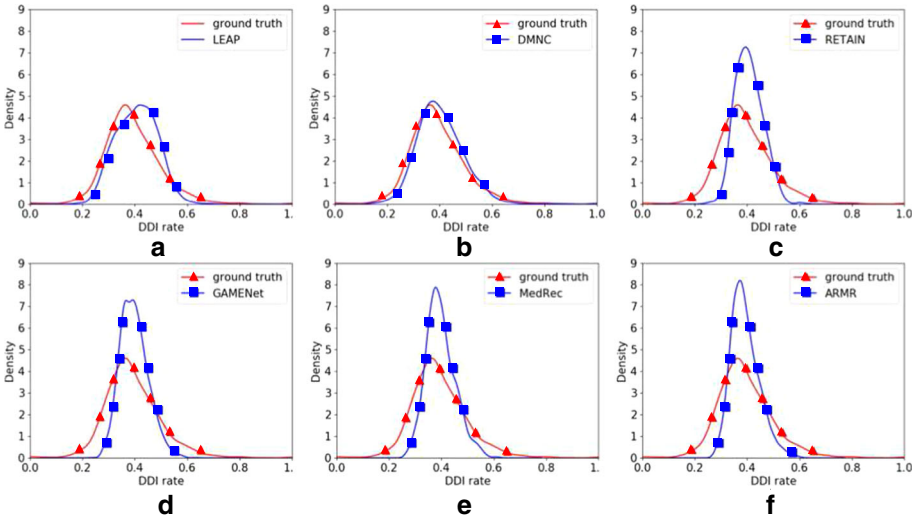


Fig. 3 Comparison of kernel estimate density on DDI rates of ground truth and recommend results

Table 4 Comparison of methods in terms of different number of DDI types. The average DDI rates for 40, 60, 80, and 100 DDI types are 0.0857, 0.1941, 0.2932, and 0.3923, respectively

#DDI types	metric	Leap	DMNC	RETAIN	GAMENet	MedRec	ARMR
40	DDI rate	0.0747	0.0959	0.0887	0.0848	0.0863	0.0813
	△DDI rate	−17.80%	8.13%	3.50%	−1.08%	0.66%	−5.17%
60	DDI rate	0.1918	0.2075	0.2166	0.2006	0.1997	0.1941
	△DDI rate	−1.19%	6.91%	11.61%	3.35%	2.87%	−0.02%
80	DDI rate	0.3099	0.3166	0.3173	0.3008	0.2955	0.2914
	△DDI rate	5.69%	7.97%	8.23%	2.62%	0.81%	−0.61%
100	DDI rate	0.4229	0.4156	0.4332	0.4043	0.3971	0.3917
	△DDI rate	7.80%	5.93%	10.43%	3.06%	1.22%	−0.15%

5 Conclusion

In this paper, we propose ARMR, a deep learning model for safe recommendation of medications. ARMR makes use of information from historical admissions by constructing a memory network and carrying out multi-hop reading on the memory network, and patient representations are regularized by a GAN model to avoid DDI. Experiments on the real-world clinical dataset MIMIC-III demonstrate that ARMR is able to achieve DDI rate reduction while ensuring effective medication recommendation. Meanwhile, there is still room for improvement. Sequential information of medications within a single admission is also important for effective medication recommendation, since those drugs that have been prescribed before would influence subsequent decisions made by doctors, so predicting medications sequentially could be a promising research direction. Also, using fitted Gaussian distribution as source of real data in the adversarial regulation may introduce bias, and a possible strategy of

future research is to use variational autoencoders (VAEs) to obtain meaningful distributions directly.

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References

1. Awad AAR, Fgaier H, Mustafa I, Elkamel A, Elnashaie SSEH (2019) Pharmacokinetic/pharmacodynamic modeling and simulation of the effect of medications on β-amyloid aggregates and cholinergic neurocycle. Comput Chem Eng 126:231–240
2. Choi E, Bahadori MT, Schuetz A, Stewart WF, Sun J (2016) Doctor AI: predicting clinical events via recurrent neural networks. In: Machine learning for healthcare conference, pp 301–318
3. Choi E, Bahadori MT, Sun J, Kulas J, Schuetz A, Stewart WF (2016) RETAIN: an interpretable predictive model for healthcare using reverse time attention mechanism. In: Advances in neural information processing systems, vol 29, Annual conference on neural information processing systems 2016, December 5-10, 2016, Barcelona, Spain, pp 3504–3512
4. Davazdahemami B, Delen D (2019) The confounding role of common diabetes medications in developing acute renal failure: a data mining approach with emphasis on drug-drug interactions. Expert Syst Appl 123:168–177
5. Ghasemi S, Etminani K, Dehghan H, Eslami S, Hasibian M, Vakili HA, Saberi M, Aghabagheri M, Namayandeh S (2019) Design and evaluation of a smart medication recommendation system for the electronic prescription. Stud Health Technol Inform 260:128–135
6. Gómez-Bombarelli R, Wei JN, Duvenaud D, Hernández-Lobato JM, Sánchez-Lengeling B, Sheberla D, Aguilera-Iparraguirre J, Hirzel TD, Adams RP, Aspuru-Guzik A (2018) Automatic chemical design using a data-driven continuous representation of molecules. ACS Cent Sci 4(2):268–276
7. Goodfellow IJ, Pouget-Abadie J, Mirza M, Xu B, Warde-Farley D, Ozair S,Courville AC, Bengio Y (2014) Generative adversarial nets. In: Advances in neural information processing systems, vol 27, Annual conference on neural information processing systems 2014, December 8-13 2014, Montreal, Quebec,Canada, pp 2672–2680
8. Graves A, Wayne G, Reynolds M, Harley T, Danihelka I, Grabska-Barwińska A, Colmenarejo SG, Grefenstette E, Ramalho T, Agapiou J et al (2016) Hybrid computing using a neural network with dynamic external memory. Nature 538(7626):471
9. Guimaraes GL, Sanchez-Lengeling B, Outeiral C, Farias PLC, Aspuru-Guzik A (2017) Objective-reinforced generative adversarial networks (organ) for sequence generation models. arXiv preprint arXiv:1705.10843
10. Guo W, Ge W, Cui L, Li H, Kong L (2019) An interpretable disease onset predictive model using crossover attention mechanism from electronic health records. IEEE Access 7:134236–134244
11. Johnson AE, Pollard TJ, Shen L, Li-wei HL, Feng M, Ghassemi M, Moody B, Szolovits P, Celi LA, Mark RG (2016) MIMIC-III, a freely accessible critical care database. Sci Data 3:160035
12. Kadurin A, Nikolenko S, Khrabrov K, Aliper A, Zhavoronkov A (2017) Drugan: an advanced generative adversarial autoencoder model for de novo generation of new molecules with desired molecular properties in silico. Mol Pharmaceut 14(9):3098–3104
13. Kumar A, Irsoy O, Ondruska P, Iyyer M, Bradbury J, Gulrajani I, Zhong V, Paulus R, Socher R (2016) Ask me anything: dynamic memory networks for natural language processing. In: International conference on machine learning, pp 1378–1387
14. Le H, Tran T, Venkatesh S (2018) Dual memory neural computer for asynchronous two-view sequential learning. In: Proceedings of the 24th ACM SIGKDD international conference on knowledge discovery & data mining. ACM, pp 1637–1645
15. Li Y, Chen W, Liu D, Zhang Z, Wu S, Liu C (2019) IFFLC: an integrated framework of feature learning and classification for multiple diagnosis codes assignment. IEEE Access 7:36810–36818
16. Lipton ZC, Kale DC, Elkan C, Wetzel RC (2016) Learning to diagnose with LSTM recurrent neural networks. In: 4th international conference on learning representations, ICLR 2016, San Juan, Puerto Rico, May 2–4, 2016, conference track proceedings

17. Ma T, Xiao C, Zhou J, Wang F (2018) Drug similarity integration through attentive multi-view graph auto-encoders. In: Proceedings of the twenty-seventh international joint conference on artificial intelligence, IJCAI 2018, July 13–19, 2018, Stockholm, Sweden, pp 3477–3483
18. Makhzani A, Shlens J, Jaitly N, Goodfellow I, Frey B (2015) Adversarial autoencoders. arXiv preprint arXiv:1511.05644
19. Malakouti S, Hauskrecht M (2019) Predicting patient's diagnoses and diagnostic categories from clinical-events in EHR data. In: Artificial intelligence in medicine—17th conference on artificial intelligence in medicine, AIME 2019, Poznan, Poland, June 26–29, 2019, Proceedings, pp 125–130
20. Mallya S, Overhage JM, Bodapati S, Srivastava N, Genc S (2019) SAVEHR: self attention vector representations for EHR based personalized chronic disease onset prediction and interpretability. arXiv preprint arXiv:1911.05370
21. McAllister-Day TK, Madhyastha TM, Lee A, Zabetian CP, Montine TJ, Grabowski TJ (2019) Effect of dopaminergic medications on blood oxygen level-dependent variability and functional connectivity in parkinson's disease and healthy aging. Brain Connect 9(7):554–565
22. Miller AH, Fisch A, Dodge J, Karimi A, Bordes A, Weston J (2016) Key-value memory networks for directly reading documents. In: Proceedings of the 2016 conference on empirical methods in natural language processing, EMNLP 2016, Austin, Texas, USA, November 1–4, 2016, pp 1400–1409
23. Nordon G, Koren G, Shalev V, Horvitz E, Radinsky K (2019) Separating wheat from chaff: Joining biomedical knowledge and patient data for repurposing medications. In: The thirty-third AAAI conference on artificial intelligence, AAAI 2019, pp 9565–9572
24. Pan S, Hu R, Long G, Jiang J, Yao L, Zhang C (2018) Adversarially regularized graph autoencoder for graph embedding. In: Proceedings of the twenty-seventh international joint conference on artificial intelligence, IJCAI 2018, July 13–19, 2018, Stockholm, Sweden, pp 2609–2615
25. Prakash A, Zhao S, Hasan S, Datla V, Lee K, Qadir A, Liu J, Farri O (2017) Condensed memory networks for clinical diagnostic inferencing. In: Thirty-first AAAI conference on artificial intelligence
26. Salimans T, Goodfellow IJ, Zaremba W, Cheung V, Radford A, Chen X (2016) Improved techniques for training gans. In: Advances in neural information processing systems, vol 29, Annual conference on neural information processing systems 2016, December 5-10, 2016, Barcelona, Spain, pp 2226–2234
27. Shang J, Ma T, Xiao C, Sun J (2019) Pre-training of graph augmented transformers for medication recommendation. In: Proceedings of the twenty-eighth international joint conference on artificial intelligence, IJCAI 2019, Macao, China, August 10–16, 2019, pp 5953–5959
28. Shang J, Xiao C, Ma T, Li H, Sun J (2019) Gamenet: graph augmented memory networks for recommending medication combination. In: Proceedings of the AAAI conference on artificial intelligence vol 33, pp 1126–1133
29. Siavelis JC, Bourdakou MM, Athanasiadis EI, Spyrou GM, Nikita KS (2016) Bioinformatics methods in drug repurposing for Alzheimer's disease. Brief Bioinform 17(2):322
30. Singh N, Halliday AC, Thomas JM, Kuznetsova OV, Baldwin R, Woon EC, Aley PK, Antoniadou I, Sharp T, Vasudevan SR et al (2013) A safe lithium mimetic for bipolar disorder. Nat Commun 4:1332
31. Spangler WS, Wilkins AD, Bachman BJ, Nagarajan M, Dayaram T, Haas PJ, Regenbogen S, Pickering CR, Comer A, Myers JN (2014) Automated hypothesis generation based on mining scientific literature. In: The 20th ACM SIGKDD international conference on knowledge discovery and data mining, KDD '14, New York, NY, USA, August 24–27, 2014, pp 1877–1886
32. Sukhbaatar S, Szlam A, Weston J, Fergus R (2015) End-to-end memory networks. In: Advances in neural information processing systems, vol 28, Annual conference on neural information processing systems 2015, December 7-12,2015, Montreal, Quebec, Canada, pp 2440–2448
33. Sun D, Ren X, Ari E, Korcsmaros T, Wu LY (2017) Discovering cooperative biomarkers for heterogeneous complex disease diagnoses. Brief Bioinform 20(1):1–13
34. Sybrandt J, Shtutman M, Safro I (2017) MOLIERE: automatic biomedical hypothesis generation system. In: Proceedings of the 23rd ACM SIGKDD international conference on knowledge discovery and data mining, Halifax, NS, Canada, August 13–17, 2017, pp 1633–1642
35. Wang M, Liu M, Liu J, Wang S, Long G, Qian B (2017) Safe medicine recommendation via medical knowledge graph embedding. arXiv preprint arXiv:1710.05980
36. Wang P, Long Z, Lv Z, Wang Z (2019) Fault detection for non-gaussian processes using multiple canonical correlation analysis models and box-cox transformation. IEEE Access 7:68707–68717
37. Wang Y, Chen W, Li B, Boots R (2019) Learning fine-grained patient similarity with dynamic bayesian network embedded RNNs. In: Database systems for advanced applications—24th international conference, DASFAA 2019, Chiang Mai, Thailand, April 22–25, 2019, Proceedings, Part I, pp 587–603
38. Wen D, Xi L, Yibo G, Lin C, Jianglong S, Di C, Kuo G, Yongshi J, Yiping Y, Jianxin C (2015) Matrix factorization-based prediction of novel drug indications by integrating genomic space. Comput Math Methods Med 2015:1–9

39. Weston J, Chopra S, Bordes A (2015) Memory networks. In: 3rd international conference on learning representations, ICLR 2015, San Diego, CA, USA, May 7–9, 2015, Conference track proceedings
40. Zhang Y, Chen R, Tang J, Stewart WF, Sun J (2017) Leap: learning to prescribe effective and safe treatment combinations for multimorbidity. In: Proceedings of the 23rd ACM SIGKDD international conference on knowledge discovery and data mining. ACM, pp 1315–1324

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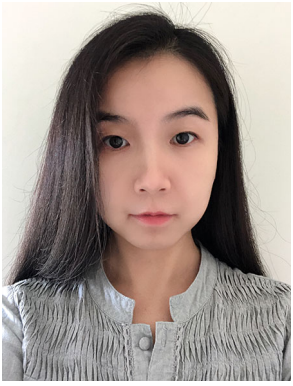
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